NHH

HOME EXAMINATION
FIE401

Fall, 2019

Start: 20.11.2019, 09:00

End: 22.11.2019, 14:00

THE HOME EXAMINATION SHOULD BE SUBMITTED IN WISEFLOW

You can find information on how to submit your paper here:
https://www.nhh.no/en/for-students/examinations/submissions-individually-groups/

Your candidate number will be announced on StudentWeb. The candidate number should be noted on all pages (not your name or student number). In case of group examinations, the candidate numbers of all group members should be noted.

SUPPLEMENTARY REGULATIONS TO HOME EXAMINATIONS

https://www.nhh.no/en/for-students/regulations/regulations-for-examinations-at-nhh-full-time-programmes/supplementary-regulations-to-the-regulations-for-examinations-at-nhh/

All pages, including front page: 5

Number of attachments: 1 (FIE401 data.zip)

FIE401 - Final Project

Formalities

This final project will be handed out on 20.11.2019 at 9:00 and has to be submitted no later than the 22.11.2019 at 14:00.

In addition to the required hand-in files, there will be an oral exam on the 28.11.2019 and 29.11.2019. Please sign up for time slots on CANVAS as a group. You do not need to be present at the presentations of your peers. You will have strictly 5 minutes to present your solution which will be followed by a question and answer session lasting up to 20 minutes. The latter will refer to both your analysis of the final project as well as all subjects covered in class.

This exam is group-based. All group members will receive the same grade for the written submission while grades on the oral exam will be individual.

Please submit the following documents

- **Coding file (.R)** Please comment your general coding approach for each task. You do not need to explain the used functions.
- **Report (.pdf)** In the report, you should present, analyse, and interpret your results. You should provide numeric and written answers to all questions asked in the exam. Do not discuss your coding in the pdf as this should be found in the R file. Please keep your answers short and precise and focus on the questions asked in the outline of the project.
- **Presentation (.pdf)** It is not allowed to change the slides between your submission and the actual presentation. We will provide your submitted presentation on the computer in the room where you give the presentation. Make sure that you use your 5 minutes presentation time wisely. It is important that you provide an economic rational for choices you made during this project.

Formal requirements

- **Report**
  - Motivate all your choices
  - Interpret your findings from a statistical and economical perspective
  - Your report should read like a mini thesis/article having following structure
    1. Abstract - summarise your findings (wordlimit = 100 words)
    2. Intro - outline the research objective
    3. Data - summarize the data and variable used (Table 1)
    4. Analysis - your analysis (Table 2 & 3)
    5. Conclusion - your findings destilled
  - The report should include three tables
    1. Summary statistics
    2. Determinants of conference call participation by institutional investors
    3. Impact of institutional investors' conference call participaton on the market
  - The maximum word count of the report without title, tables, abstract, and header is 2500.
  - Make sure that tables are self-contained and as described in the last lecture
  - Present a maximum of six models in each of Table 2 & 3
  - Use fairly plain formatting ensuring readability
  - No appendix

  - No figures
  - You do not need to explain the empirical models used as this will be part of the oral exam
- **Presentation**
  - The main purpose of the presentation is to be a starting point for the oral exam
  - The time limit of 5 minutes will be strictly enforced
  - Discuss your choices of how you set up the econometric analysis
  - All three tables of the report have to be in the presentation
  - You do not need to prepare an appendix for the presentation
  - Do not write your names or any identifying information on the slides

Your task

Conference calls are held almost each quarter at the time a firm releases their quarterly results. These conference calls are an opportunity for market participants to gather information from management additional to what is published officially. Please see following link https://seekingalpha.com/article/4298445-tesla-inc-tsla-ceo-elon-musk-q3-2019-results-earnings-call-transcript for a sample of an earnings call transcript.

The main group of participants in conference calls are sell-side analysts, referring to analysts that sell analysis such as stock recommendations but do not invest by themselves or through the fund they work for.

Another group of conference call participants are institutional investors such as the Vanguard Group or Blackrock but also pension funds such as the Norwegian Oil Fund. The main difference between institutional investors and sell-side analysts is that institutional investors actively invest and that they are the biggest players in the market these days. In addition, they often actively engage in governance. For the purpose of this exam, we can think of governance as all corrective actions by owners of a firm. For example, if an institutional investors disagrees with a proposed firm strategy he could engage in several governance related activities such as private discussions with management, voting against management, or going to conference calls signalling their discontent to the market.

Hence, if institutional investors attend a conference call, this might provide new information to other investors, and the market might react. If the market did not expect that institutional investors attend, this might cause uncertainty and some investors might adjust their position which leads to abnormal trading volume. Institutional investors' participation in conference calls might be a negative signal - maybe they disagree with management - or a positive signal - they stick with the firm and do not sell their shares given a negative event. Hence, whether and how the market reacts is an empirical question.

However, maybe there is no information in the observation that an institutional investor participates in a conference calls.

In this exam, you are investigating following research questions:

Does conference call participation by institutional investors affect the market?

Given the supplied data, you will have to settle for a set of dependent and independent variables. Make sure that your choices are motivated by economic intuition spelled out in your report.

Your analysis proceeds in three steps (each step is a table):
1. Summarise the data and variables used
2. Explain conference call participation by institutional investors with observable factors
3. Investigate a link between institutional investors' conference call participation and the market

There are several reasons why institutional investors may attend a conference call of a firm related to the firm's behaviour and performance. For example, if a firm experiences unexpected negative earnings, engages in diversifying mergers, the CEO engages in value destroying activities, the institution disagrees with the strategy, etc. institutional investors might decide to attend the conference call. In general, institutional investors will most likely only attend conference calls of firms they are invested in.

There might be other reasons of why institutional investors may or may not attend a conference call of a firm unrelated to the firm in question. For example, institutional investors might be distracted if there are many other conference calls or merger activity on the same day or within the same industry. In addition, institutional investors often hold stocks in the same industry. So, if the industry experiences a negative return shock, they might focus on another firm than the one holding the call.

Additional guidance

- Please use information and data as provided in the exam. In other words, do not gather additional data.
- Using code found on the internet is allowed, as long as you reference the sources correctly.
- Plagiarism in any way is not allowed.
- Do not be discouraged if data is fairly messy, it usually is.
- Try to understand the data before you decide on an econometric approach.
- You need to make informed and motivated choices for variables and models used. Use economic intuition rather than maximizing statistical significance or R2.
- Make sure that your results are not driven by outliers in your suggested variables.
- Make sure that you decide on variables and data sets you want to use in your model, before starting to code.
- If you have questions during the exam, do not visit my office but write an email. If I answer your request, I will publish my comment on Canvas so that information is public and distributed fairly.
- We encourage students facing problems with R to utilize the Virtual Desktop Infrastructure that NHH offers or computers in the library.
- Good luck!

Data

1. `calls.txt` is a data.frame that includes information on individual conference calls. Following variables are included:
   - `date`, `hour`, `minute`: timing of the conference call
   - `cusip`, `lpermno`, `gvkey`: firm identifiers
   - `institution_participates`: Indicator variable turning one if at least one institution participated in the conference call, and zero otherwise
   - `participants`: number of conference call participants
   - `mktv_end_of_last_month` and `prc_end_of_last_month`: Market value (in thousands) and price of the firm at the end of the month prior to the conference call.
   - `SUE`: Standardized Unexpected Earnings is a standardised measure of how surprising the current earnings are. Positive indicates that the earnings are higher than expected.
   - `sic`: SIC codes are four-digit numerical representations of industries
   - `FF.49`: industry classification by Fama and French, please see Kenneth French's webpage for more details.
   - A set of variables related to the day surrounding the conference call
     - Variable type (the beginning of the variable name): HML indicates the return on the HML factor, SMB indicates the return of the SMB factor, $MKT.RF$ indicates return on the market minus the risk free rate, RET indicates the return of the stock in excess of the risk free rate,

and VOL is the total number of shares of a stock sold. All return variables are not converted to percentages (you need to multiply with 100 to get percentages).
    - Timing (the end of the variable name): $l2$ indicates 2 days before the call, $l1$ indicates 1 day before the call, $t0$ indicates the day of the call, $f1$ indicates 1 day after the call, $f2$ indicates 2 days after the call
    - This combines into a variable in following way: e.g. `RET.f2` is the excess return of the stock two days after the conference call
  - A set of variables computed over an extended time before the conference call
    - Variable type (the beginning of the variable name): `beta_XXX` is the `MKT`, `HML`, or `SMB` (substitute for `XXX`) beta of the firm computed given the respective window. `av_daily_return` is the average excess return of the stock given the respective window. `av_daily_vol` is the average daily volume given the respective window.
    - Timing (the end of the variable name): `short_window` refers to the window of [-60:-3] trading days before the day of the call. `long_window` refers to the window of [-250:-3] trading days before the day of the call.
    - This combines into a variable in following way: e.g. `av_daily_vol_short_window` is the average excess return of the stock computed using the time window of [-60:-3] trading days before the day of the call.
2. `compustat.csv` is a data.fame that includes quarterly accounting data of firms. Following variables are included:
  - `announcement.date`: day the financial information is announced
  - `lpermno`, `gvkey`: firm identifiers
  - `fiscal.quarter`: indicates which fiscal quarter the financial data refers to
  - `fiscal.year`: indicates which fiscal year the financial data refers to
  - `assets`: assets of a firm expressed in millions
  - `income`: income of a firm expressed in millions
  - `sales`: sales of a firm expressed in millions
  - A comment on merging: the financial data is usually announced in the same month of the conference call (hint: make a variable capturing year and month of the announcement date). If you use `left_join()` for merging the data, make sure that the variable names you want to merge with as well as the class of the merging variables is the same. It is easiest to use `substr()` to filter year and month from a date variable than learning how to do date conversions if you never used them.
3. `industry.returns.daily.Rdata` is a data.frame that includes daily returns for individual `FF.49` industries
  - `date`: date of the return
  - All other columns hold returns for individual industries as indicated by the column names in percent. Note that the column names match perfectly with `calls$FF.49`
  - If you choose to use this data.frame, you might find the function `match()` useful
  - Another approach would be to transform `industry.returns.daily` and use `left_join()` given the right variable names for merging
4. `merger.csv` is a data.frame that includes all mergers in the sample period
  - `lpermno`, `gvkey`: firm identifiers
  - `announcement.date`: announcement day of the merger
  - `return.runup`: return runnup prior to the merger in percent
  - `hostile`: indicator if the merger was hostile
  - `public`: indicator if the target was public
  - `private`: indicator if the target was private
  - `all.cash`: indicator if the deal was paid in cash
  - `all.stock`: indicator if the deal was paid in stock
  - `relative.size`: size of the target relative to the buyer
  - `total.size`: size of the target